

Practical - 3

Aim: There are N children standing in a line with some rating value. You want to distribute a minimum number of candies to these children such that: Each child must have at least one candy. The children with higher ratings will have more candies than their neighbor's. You need to write a program to calculate the minimum candies you must give.

Source Code:

```
public class CandyDistribution {  
  
    public static int distributeCandies(int[] ratings) {int n =  
        ratings.length;  
        int[] candies = new int[n];for  
        (int i = 0; i < n; i++) {  
            candies[i] = 1;  
        }  
        for (int i = 1; i < n; i++) {  
            if (ratings[i] > ratings[i - 1]) {  
                candies[i] = candies[i - 1] + 1;  
            }  
        }  
        for (int i = n - 2; i >= 0; i--) {  
            if (ratings[i] > ratings[i + 1]) {  
                candies[i] = Math.max(candies[i], candies[i + 1] + 1);  
            }  
        }  
        int totalCandies = 0;  
        for (int candy : candies) {  
            totalCandies += candy;  
        }  
        return totalCandies;  
    }  
    public static void main(String[] args) {int[]  
        ratings = {3, 1, 5, 2, 4};  
        int minCandies = distributeCandies(ratings); System.out.println("Minimum candies  
        required: " + minCandies);  
    }  
}
```

Output:

CandyDistribution.java

NEW

JAVA

RUN

```

1 public class CandyDistribution {
2
3 public static int distributeCandies(int[] ratings) { int n = ratings.length;
4 int[] candies = new int[n]; for (int i = 0; i < n; i++) {
5 candies[i] = 1;
6 }
7 for (int i = 1; i < n; i++) {
8 if (ratings[i] > ratings[i - 1]) { candies[i] = candies[i - 1] + 1;
9 }
10 }
11 for (int i = n - 2; i >= 0; i--) {
12 if (ratings[i] > ratings[i + 1]) {
13 candies[i] = Math.max(candies[i], candies[i + 1] + 1);
14 }
15 }
16 int totalCandies = 0;
17 for (int candy : candies) { totalCandies += candy;
18 }
19 return totalCandies;
20 }
21 public static void main(String[] args) { int[] ratings = {3, 1, 5, 2, 4};
22 int minCandies = distributeCandies(ratings); System.out.println("Minimum candies required:
23 }
24 }

```

STDIN

Input for the program { Optional }

Output:

Minimum candies required: 8